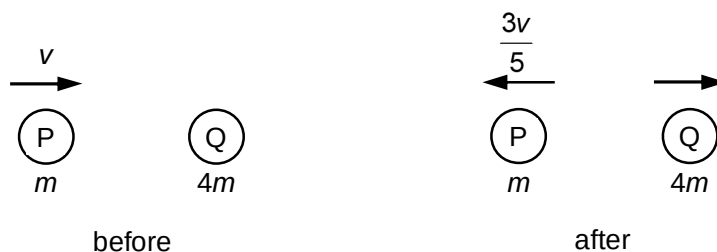


- 4 An elastic collision occurs between two bodies P and Q. The mass of body P is m and the mass of body Q is $4m$. Body P travels at speed v before the collision and speed $\frac{3v}{5}$ in the opposite direction after the collision. Body Q is stationary before the collision.



What is the kinetic energy of body Q after the collision?

- A $\frac{1}{5}mv^2$
- B $\frac{8}{25}mv^2$
- C $\frac{17}{25}mv^2$
- D $\frac{4}{5}mv^2$

- 5** A door is made up of a uniform pane of mass 20 kg. The pane is 0.50 m wide and 0.50 m